COMPUTATIONAL INVESTIGATION ON AERODYNAMIC PERFORMANCE OF ZIG-ZAG SURFACE ROUGHNESS OVER AIRFOIL

AN INTERNSHIP PROJECT REPORT

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This is to certify that the following students underwent a **Summer Internship Programme** in the **Department of Aerospace Engineering, School of Mechanical Engineering, SRM Institute of Science and Technology (SRMIST), Kattankulathur**, during the period from **01.06.2026 to 30.06.2026**.

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This project report entitled **"Computational Investigation on Aerodynamic Performance of Zig-Zag Surface Roughness over Airfoil"** is the original work carried out by the above student under the guidance of **Prof. S. Senthilkumar, Ph.D.** Department of Aerospace Engineering, SRM Institute of Science and Technology, during the internship period. The work embodied in this report has not been submitted elsewhere for the award of any degree, diploma, or certificate.

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ABSTRACT

Surface roughness can be used as a passive flow control technique to modify boundary layer behaviour and influence flow separation when introduced at suitable chordwise locations. This project investigates the effect of different surface roughness positions on the aerodynamic performance of a NACA 4412 airfoil using Computational Fluid Dynamics (CFD).

A two-dimensional NACA 4412 airfoil was modelled and analysed in ANSYS Fluent. Geometrical modifications were carried out by introducing surface roughness at 20%, 30%, 40%, 50%, and 60% of the chord length, in addition to the smooth baseline airfoil. CFD simulations were performed for angles of attack ranging from 0° to 25°. The aerodynamic performance of each configuration was evaluated by comparing the lift coefficient (CL), drag coefficient (CD), and aerodynamic efficiency (CL/CD). Based on these comparisons, the optimum roughness configuration was identified, after which a detailed investigation of the velocity and pressure contours was carried out by comparing the smooth airfoil with the 40% roughness configuration at each angle of attack to examine changes in flow separation and pressure distribution.

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List of Symbols

Symbol	Description	Unit
Cl	Lift coefficient	-
CD	Drag coefficient	-
Cl/Cd	Lift-to-drag ratio	-
α	Angle of attack	^{o}
Re	Reynolds number	-
V	Freestream velocity	m/s
P	Static pressure	Pa
ρ	Density of air	kg/m^3
μ	Dynamic viscosity	Kg/(m.s)
c	Airfoil chord length	m
h	Zig-zag roughness height	mm
CFD	Computational Fluid Dynamics	-
NACA	National Advisory Committee for Aeronautics	-

CHAPTER 1
INTRODUCTION

1.1 Overview of Airfoil

An airfoil is a specially designed aerodynamic surface that creates lift as air moves over it. Airfoils are found in many applications including wings on aircraft, rotor blades for helicopters and wind turbines. In all these applications, the performance of the airfoil affects the performance of the aircraft/wind turbines. There are several different airfoils that can be used in such applications and one such common airfoil that is used in aerodynamic studies includes NACA airfoils since their geometries and performance characteristics have been well documented. One of the most studied NACA series airfoils includes NACA 4412 due to the good lift characteristics it possesses [1]. The performance of airfoils is usually measured using parameters like lift and drag coefficients which measure the ability of the airfoil to create lift and reduce aerodynamic losses respectively. In wind turbines, this becomes even more important as the behavior of the blade section determines how much energy can be extracted from the wind [5] .

There is a close relationship between the aerodynamics performance of the airfoil and the manner in which the fluid flows over the airfoil. In the process of flowing over the airfoil, a fluid film that is referred to as the boundary layer forms on the surface of the airfoil. The nature of this boundary layer is an important determinant of the formation of lift, drag and the general aerodynamic performance of the airfoil. *At low angles of attacks, the flow is usually attached to the surface of the airfoil*, hence the good aerodynamics performance. However, when the angle of attack is increased, there is an adverse pressure gradient created within the flow which causes the flow to separate from the surface of the airfoil. Flow separation causes deterioration in lift and an increase in drag. *Separation of flow is one of the factors that causes the airfoil to stall*. There is another factor worth mentioning in this context, and this is the transition of the boundary layer from laminar to turbulent flow.

Prior research has greatly expanded the existing knowledge regarding the aerodynamics of airfoils with varying angles of attack and surface roughness. The

research done with leading-edge contamination, sandpaper roughness, and zig-zag roughness strips has revealed that the surface roughness is one of the key determinants of lift, drag, boundary layer transition, and stall [1] , [5] , [6] , [7]. Apart from that, there are a number of investigations which focus on the effect of roughness location on wind turbine airfoils and other aerodynamic surfaces [2] , [7]. Still, additional research is needed in order to clarify the way in which the location of zig-zag roughness affects the aerodynamics of NACA 4412 airfoil. In particular, the study of the influence of zig-zag roughness located at the distance of 20%, 30%, 40%, 50%, and 60% chord will help to obtain new information on the impact of roughness on lift coefficient, drag coefficient, and stall.

1.2 Surface Roughness

A number of researchers have studied the effects of surface roughness on the aerodynamic performance of airfoils under various roughness configurations and operational conditions[5] , [6]. Research on sandpaper roughness, surface contamination roughness models and zig-zag roughness strips have shown significant changes in the aerodynamic properties of airfoils when compared to smooth airfoils [6]. Generally, the introduction of roughness causes differences in the boundary layer formation process, leading to lift and drag coefficient changes. Most research papers have indicated the reduction in the value of the maximum lift coefficient and poor lift-to-drag ratio as a result of the early transition of the boundary layer and pressure changes around the airfoil surface. Several previous studies on wind turbines with zig-zag roughness have shown that the location of roughness has a significant effect on the aerodynamic performance of the airfoil where roughness close to the leading edge is more effective than the one away from the leading edge. Besides lift and drag property changes, several other studies have shown changes in the stall behaviour as a result of boundary layer transition and flow separation induced by roughness.

Among the various methods used to simulate surface roughness, zig-zag roughness elements have received considerable attention in aerodynamic and wind turbine studies [2]. Unlike random surface contamination, zig-zag roughness is intentionally introduced to study its influence on boundary layer behaviour and aerodynamic performance [7]. The primary function of a zig-zag roughness strip is to promote an earlier transition from laminar flow to turbulent flow at a desired location

on the airfoil surface. By triggering this transition, the boundary layer gains additional momentum and its resistance to flow separation may be altered. Because of these characteristics, zig-zag roughness has been widely used in wind tunnel experiments and numerical investigations involving wind turbine airfoils. Previous studies have shown that the aerodynamic response of an airfoil is influenced not only by the size of the roughness element but also by its location along the chord. As a result, zig-zag roughness has become a useful tool for examining the relationship between boundary layer transition and aerodynamic performance.

Many researches have been done about the aerodynamic properties of airfoils depending on their angles of attack, while many others have studied the impact of roughness on the surfaces of airfoils with different roughness patterns. Nevertheless, there has not been much attention paid to the impacts of roughness location on the aerodynamic performance of the NACA 4412 airfoil. Most of the existing researches deal with the impact of leading edge contamination, sandpaper roughness or zig-zag strips on the aerodynamic performance of wind turbine airfoils. On the contrary, fewer researches study the behavior of NACA 4412 with different roughness patterns. In particular, there is no enough research done regarding the impact of zig-zag roughness at different chord positions on the lift, drag and stall of the airfoil. Moreover, there is very little knowledge about the impact of roughness placed at 20%, 30%, 40%, 50% and 60% chord on the aerodynamic performance of NACA 4412 depending on the angle of attack **[1]**.

The current research seeks to understand the impact of zig-zag surface roughness on the aerodynamic performance of the NACA 4412 airfoil by employing Computational Fluid Dynamics (CFD). A smooth airfoil geometry is firstly examined and verified against available literature data **[1]**. Surface roughness elements in the form of zig-zags with a height equal to 0.8% of the chord are applied to the airfoil surface at 20%, 30%, 40%, 50% and 60% of the chord positions. Simulations will be carried out at angles of attack of 0°, 5°, 10°, 15° and 20° to assess the changes in lift coefficient, drag coefficient and general aerodynamic performance. Special emphasis will be placed on the role of surface roughness location on boundary layer formation and stall characteristics. The results of this research will hopefully provide further insight into the impact of surface roughness on aerodynamic performance.

CHAPTER 2
METHODOLOGY

2.1 Design Configuration

In this current study, the effect of zig-zag surface roughness on the aerodynamic performance of NACA 4412 airfoil is studied with the help of Computational Fluid Dynamics (CFD). NACA 4412 airfoil has been chosen for the experiment due to its wide application in aerodynamics and wind turbines along with its well-documented aerodynamic properties. This airfoil has a chord length of 0.1m and maximum camber of 4% at 40% chord length and a maximum thickness of 12% at 30% chord length. Coordinates of this airfoil were taken from Airfoil Tools website and its geometry was prepared with the help of CATIA software.

In order to model the roughness geometry, the zig-zag roughness geometry was used based on earlier studies on airfoil of wind turbines. *The roughness geometry was made up of series of equilateral triangles having an angle of 60°*. Each triangle had a *width of 1 mm and height of 0.8 mm,* which corresponds to 0.8% of the chord length. Five roughness geometries apart from the smooth baseline airfoil were studied. In the case of 20% roughness, the roughness was placed from the region 60% of chord length. The 30% roughness geometry covered from the region 50% of chord length, whereas the 40% roughness geometry covered from the region 60% of chord length. In the case of 50% roughness, the roughness covered the rear half of the airfoil from 50% of the chord length to the trailing edge, whereas the 60% roughness geometry covered the region from 40% of the chord length to the trailing edge.

The geometry of the external flow was constructed around the airfoil, and the *C-grid topology* was used as the grid topology of the computational domain. An upstream distance of 1 m and downstream distance of 2 m were maintained from the airfoil. The distances of the top and bottom walls from the airfoil were maintained as 1 m in order to minimize the influence of the boundary effect on the solution. A C-grid domain was adopted for the present simulation to ensure sufficient space for flow development around the NACA 4412 airfoil at a Reynolds number of 2.9×10^5. The airfoil was positioned away from the inlet, outlet, and far-field boundaries so that

boundary effects would not influence the pressure distribution and aerodynamic coefficients. A longer downstream region was maintained to capture wake development and possible flow separation at higher angles of attack. The dimensions of the computational domain were chosen after considering the chord length and expected flow behaviour around the airfoil, while keeping the computational cost within reasonable limits.

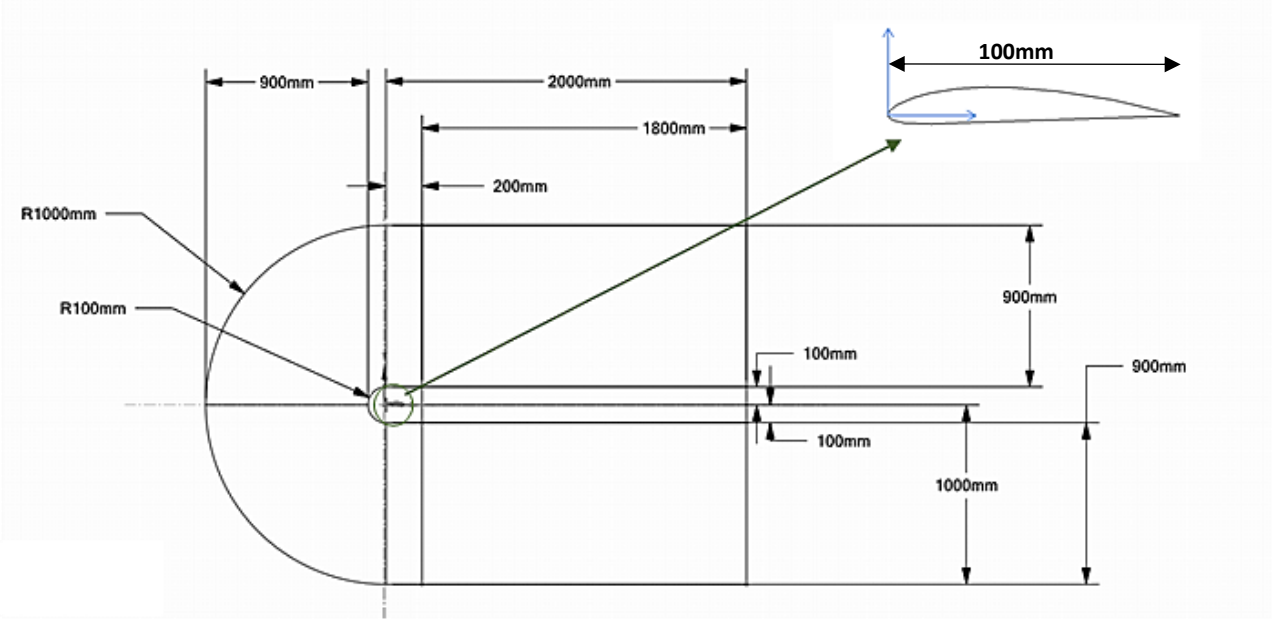


Figure 2.1- "C" Domain Sizing

2.2 Mesh Generation

Mesh generation was done using ANSYS Fluent Meshing in ANSYS Workbench. A structured mesh was created for the entire domain along with the refining of the mesh close to the airfoil. For accurate representation of the boundary layer growth, *20 inflation layers with the growth ratio of 1.2 and the first layer thickness of 1.4055 x* 10^{-5} *m* were provided around the airfoil. Mesh refinement was concentrated in regions where strong flow gradients were expected, particularly near the leading edge, trailing edge, and wake. Inflation layers were generated along the airfoil surface to resolve the boundary layer and improve the prediction of near-wall flow behaviour. Since the present work investigates the influence of surface roughness, accurate boundary layer resolution was necessary because roughness directly affects transition, pressure distribution, and separation characteristics. Away from the airfoil, a progressively coarser mesh was employed as the flow gradients became less significant.

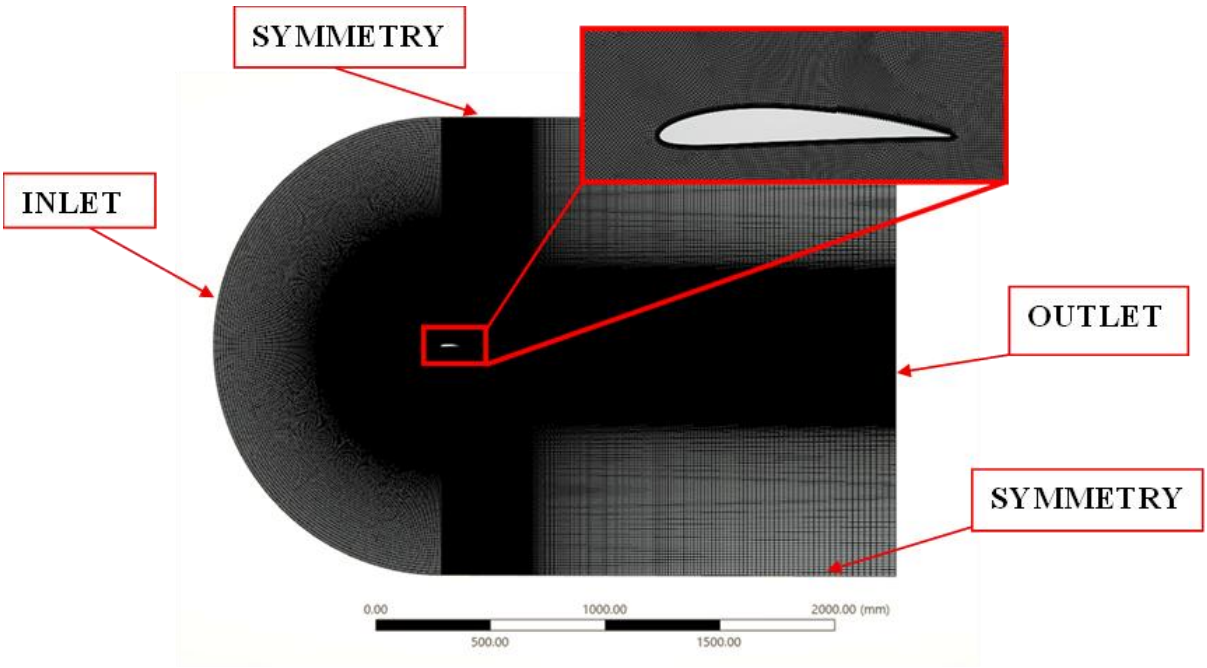


Figure 2.2: Sample Mesh of 40% Roughness Airfoil

2.3 Setup Parameters

Table 2.1: Table of Parameters

Parameters	Values
Fluid	Air
Velocity	43 m/s
Density	1.225 kg/m^3
Angle of Attack	0°, 5°, 10°, 15°, 20°, 25°
Chord Length	0.1 m
Momentum	Second Order
Viscosity	1.7894×10^{-5} kg/m·s
Number of Iteration	2000

Numerical simulations were carried out by means of ANSYS Fluent 2025 R2 software. As a working fluid, *air with the density of 1.225 kg/m³ and the dynamic viscosity of* 1.7894×10^{-5} *kg/m·s was chosen. An x-directional velocity inlet of 43 m/s was applied to the inlet, giving the Reynolds number of* 2.94×10^5*, based on the chord length of the airfoil*. The pressure *outlet with zero-gauge pressure* was used as the outlet boundary, whereas symmetry conditions were assigned to the upper and lower

boundaries. *The airfoil surface was defined as a no-slip stationary wall.* A pressure-based transient solver together with the coupled method of pressure-velocity coupling were used. Transition k-kl-ω turbulence model was applied in most of the cases, while k-ω SST turbulence model was used for simulations with an angle of attack in excess of 20 degrees in order to provide better predictions of flow separation behavior under stall conditions. The time-step size of 0.1 s and 1000time steps were used for each simulation. The criterion of convergence for all residuals was set to 10^{-6} with the maximum number of iterations equal to 2000.

Before investigating the effects of the roughness cases, the computational fluid dynamics model was validated through performing simulations with a smooth NACA 4412 airfoil. Comparison of the results obtained from the simulations in terms of lift coefficient at different angles of attack of 0°, 5°, 10°, 15° and 20° with reference data proved acceptable accuracy in agreement with the numerical and published results[1]. The comparison of results confirmed the validity of the numerical approach to be used in this study. Following the validation process, simulations were performed for the smooth airfoil and roughness cases at 0°, 5°, 10°, 15° and 20° angles of attack. In total, 30 simulations were performed which included five simulations of the baseline case and 25 roughness cases. The lift coefficient (Cl) and drag coefficient (Cd) were used to investigate aerodynamic performance. Velocity and pressure contours were also used to investigate flow behaviour and the effects of roughness on separation an

stall behaviour.

CHAPTER 3
RESULTS AND DISCUSSION

3.1 Validation

The aerodynamic performance of the baseline NACA 4412 airfoil and the five zig-zag roughness configurations was evaluated using the lift coefficient (Cl) and drag coefficient (Cd) over an angle-of-attack range of 0° to 25°. Prior to analysing the roughness configurations, the numerical model was verified against published aerodynamic data for the baseline airfoil under similar operating conditions. The agreement obtained between the numerical predictions and the reference results provided sufficient confidence to extend the investigation to different roughness locations. The following discussion focuses on the changes in aerodynamic performance caused by positioning the zig-zag roughness at 20%, 30%, 40%, 50% and 60% of the chord.

3.1.1 Design Pattern for Surface Validation

To provide a consistent basis for comparison, two airfoil configurations were selected for the contour analysis. The first configuration represents the baseline NACA 4412 airfoil with a smooth surface, as illustrated in **Figure 3.1**. The second configuration, shown in **Figure 3.2**, incorporates a zig-zag roughness element positioned at **40% of the chord length** along the upper (suction) surface of the airfoil. This location was selected based on the aerodynamic coefficient analysis presented in the previous section, where it demonstrated the most favourable overall performance among the roughness configurations investigated. Using these two models under identical boundary conditions allows the influence of the zig-zag roughness on the flow field to be examined directly through pressure and velocity contour comparisons, while keeping all other simulation parameters unchanged.

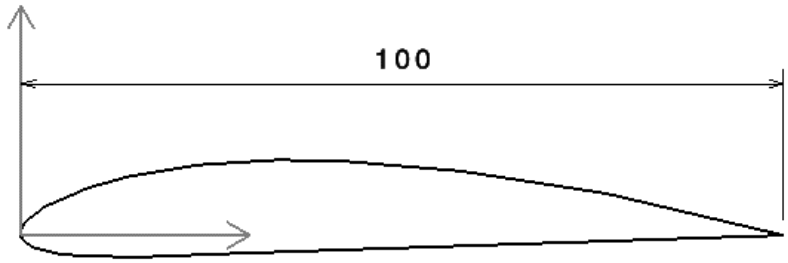


Figure 3.1 Baseline NACA 4412 airfoil used for contour comparison

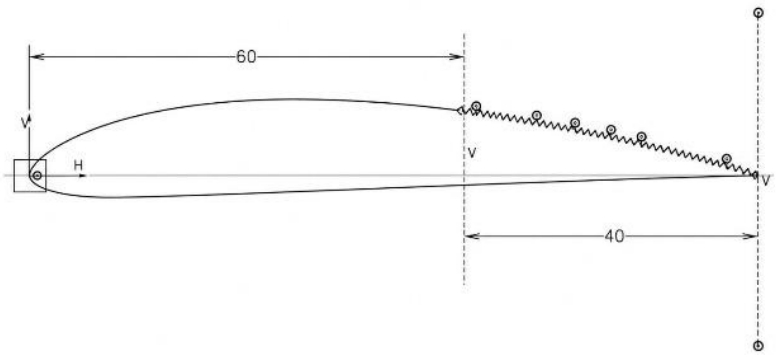


Figure 3.2 NACA 4412 airfoil with zig-zag roughness located at 40% chord used for contour comparison.

Table 3.1- Coefficient of lift at different Angles of Attack with Different Roughness

Angle of Attack (º)	Reference Cl	Baseline Cl	ROUGHNESS				
			Cl (20%)	Cl (30%)	Cl (40%)	Cl (50%)	Cl (60%)
0	0.41	0.46	0.27	0.25	0.23	0.23	0.22
5	0.91	0.74	0.55	0.51	0.51	0.5	0.49
10	1.26	1.00	0.79	0.79	0.75	0.77	0.75
15	1.24	1.21	1	0.98	1.01	0.99	0.99
20	0.91	0.95	1.06	1	1.09	0.98	0.99
25	-	0.90	0.96	0.98	1.04	1.3	1.1

Table 3.2-Coefficient of Drag at different Angles of Attack with Different Roughness

Angle of Attack (º)	Reference Cd	Baseline Cd	ROUGHNESS				
			Cd (20%)	Cd (30%)	Cd (40%)	Cd (50%)	Cd (60%)
0	0.01	0.01	0.04	0.04	0.03	0.04	0.04
5	0.04	0.04	0.06	0.07	0.06	0.07	0.07
10	0.06	0.09	0.11	0.1	0.1	0.11	0.11
15	0.1	0.16	0.17	0.17	0.16	0.17	0.11
20	0.2	0.2	0.2	0.21	0.21	0.21	0.21
25	-	0.2	0.25	0.28	0.26	0.24	0.27

The numerical results reveal that the influence of roughness location becomes increasingly apparent as the angle of attack increases. At lower angles of attack, the lift coefficients remain relatively close for all configurations, indicating that the flow remains largely attached and the effect of the roughness element on the boundary layer is limited. As the airfoil approaches higher incidence, the differences between the configurations become more pronounced. Variations in roughness position alter the development of the boundary layer, producing noticeable changes in both lift generation and aerodynamic resistance. These differences are reflected in the aerodynamic coefficients and become most evident near the onset of flow separation.

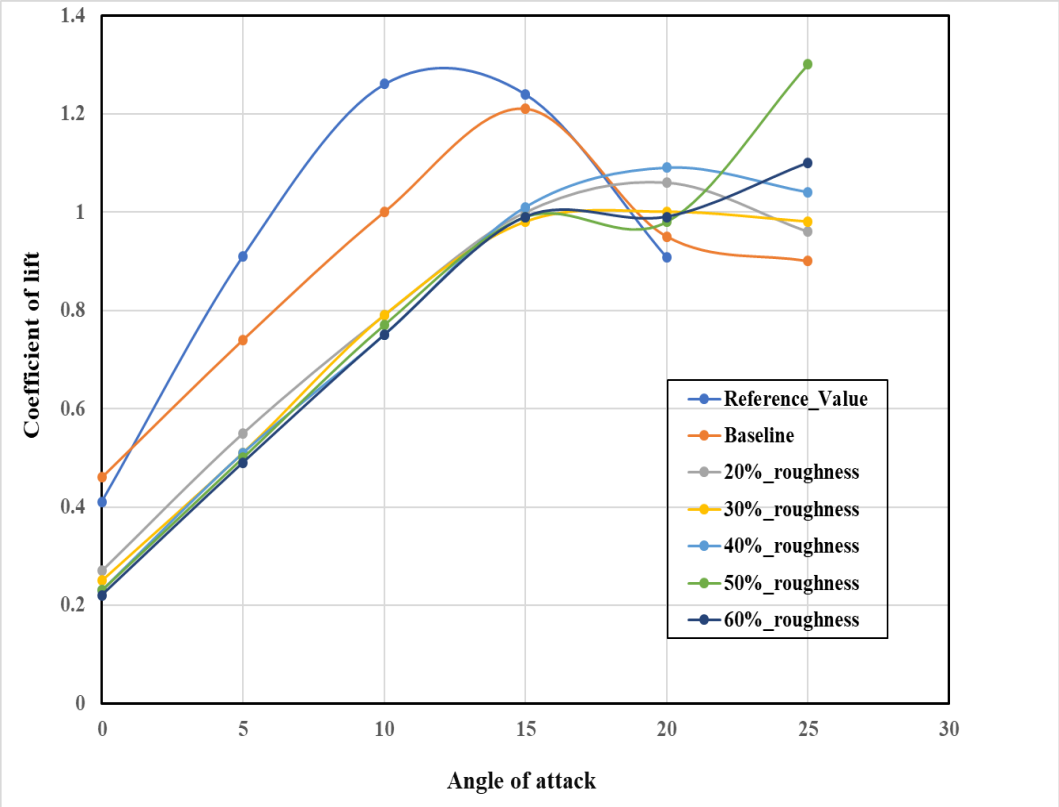


Figure 3.3 Variation of lift coefficient with angle of attack for the baseline airfoil and all roughness configurations

The lift coefficient increases progressively from 0° to approximately 15° for every configuration before reducing at higher angles of attack. This behaviour is consistent with the increase in pressure difference between the upper and lower surfaces as the airfoil is inclined into the freestream, followed by the *gradual loss of lift associated with boundary-layer separation.* While all roughness configurations follow the same overall trend, the magnitude of the lift coefficient differs once the angle of attack exceeds the moderate operating range. Roughness located further downstream generally maintains lift values closer to the baseline at higher angles of attack, whereas configurations positioned nearer the leading edge experience larger deviations. The difference suggests that the interaction between the roughness element and the developing boundary layer depends strongly on its chordwise location, particularly as adverse pressure gradients become more dominant.

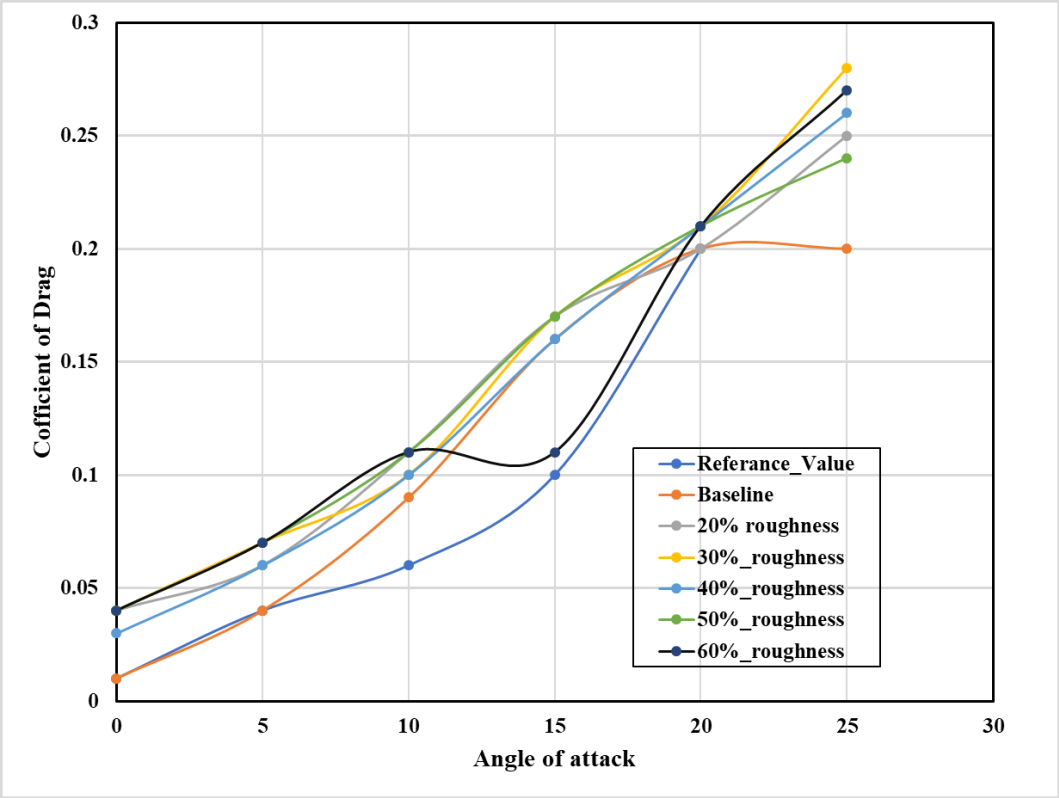


Figure 3.4 Variation of drag coefficient with angle of attack for the baseline airfoil and all roughness configurations.

The drag coefficient remains comparatively low at small angles of attack and increases steadily as the airfoil approaches stall. This increase is expected because pressure drag becomes progressively more significant as the separated flow region grows over the suction surface. Although every roughness configuration follows the same general trend, the rate of increase varies with roughness location. Differences that are almost negligible at lower angles of attack become increasingly noticeable beyond 10°, where the combined effects of boundary-layer development and flow separation begin to influence the aerodynamic behaviour. The variation in drag demonstrates that the position of the zig-zag roughness affects not only lift generation but also the overall aerodynamic losses experienced by the airfoil.

Taken together, the lift and drag characteristics show that relocating the zig-zag roughness along the suction surface changes the aerodynamic response of the NACA 4412 airfoil, particularly at moderate and high angles of attack. Since these coefficients represent the overall aerodynamic performance, a detailed examination of the flow field

is required to explain the observed trends. The pressure and velocity contour analyses presented in the following sections are used to identify the flow mechanisms responsible for the differences between the investigated roughness configurations.

3.2 Flow Field Analysis

The velocity contours provide a detailed representation of the flow development around the airfoil and illustrate how the introduction of zig-zag roughness influences the surrounding flow field. A comparison between the baseline NACA 4412 airfoil and the 40% chord roughness configuration was carried out at angles of attack ranging from 0° to 25°. Since both models were analysed under identical boundary conditions, the observed differences in the velocity field are associated with the presence of the roughness element and its interaction with the developing boundary layer. The following discussion examines the flow behaviour at each angle of attack and relates the changes in the velocity distribution to the aerodynamic performance presented in the previous section.

The velocity contours are interpreted together with the aerodynamic coefficients obtained in Section 3.1 to explain the influence of roughness on flow acceleration, wake development, and the onset of flow separation. This comparison provides a clearer understanding of the flow mechanisms responsible for the observed aerodynamic behaviour.

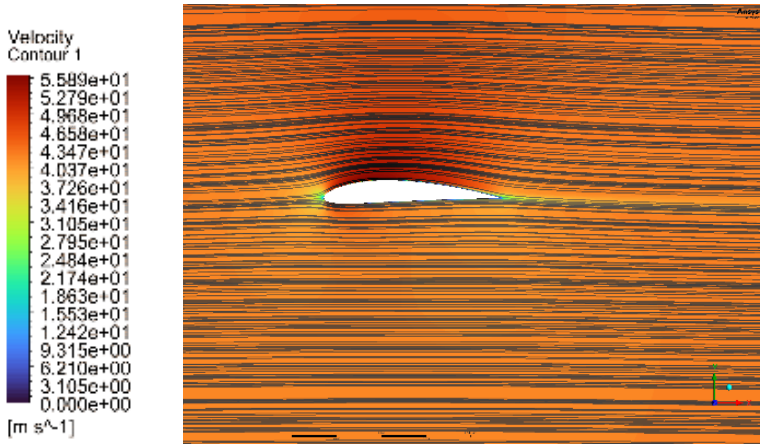


Figure 3.5 Velocity contour of baseline airfoil at angle of attack 0

At an angle of attack of 0°, the airflow remains attached over the entire airfoil surface for both configurations, indicating stable flow conditions with no evidence of separation. The velocity distribution around the baseline airfoil shows a gradual acceleration of the flow over the upper surface as the streamlines follow the cambered profile, while the flow beneath the airfoil experiences comparatively smaller changes in velocity. This velocity difference between the upper and lower surfaces initiates the pressure difference responsible for lift generation, even at zero angle of attack.

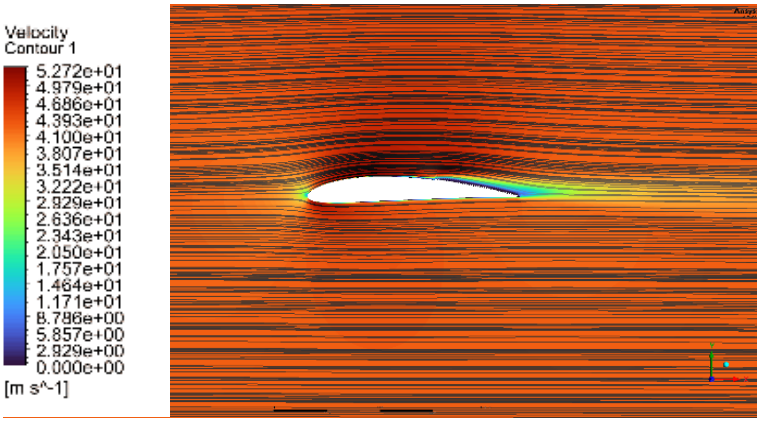


Figure 3.6 Velocity contour of 40% roughness airfoil at angle of attack 0

Introducing the zig- a zag roughness at 40% of the chord produces only localized modification to the velocity field. A slight disturbance develops near the roughness location as the boundary layer interacts with the surface irregularity, but the surrounding flow quickly recovers downstream. The overall velocity distribution remains close to that of the baseline configuration, indicating that the roughness has only a minor influence while the boundary layer is still fully attached..

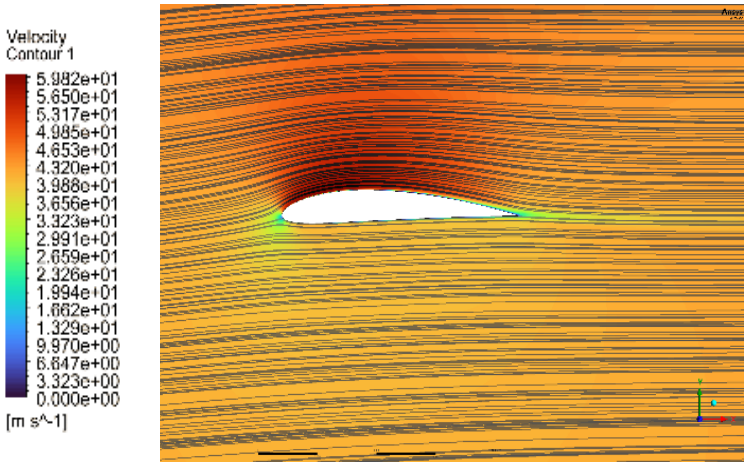


Figure 3.7 Velocity contour of baseline airfoil at angle of attack 5

As the angle of attack increases to 5°, the airflow accelerates further over the upper surface of both airfoils, producing a larger high-velocity region than that observed at 0°. The increase in flow velocity over the suction surface indicates a stronger pressure difference across the airfoil, contributing to the corresponding increase in lift coefficient. The flow remains attached throughout the airfoil surface, and no significant flow separation is observed under these operating conditions.

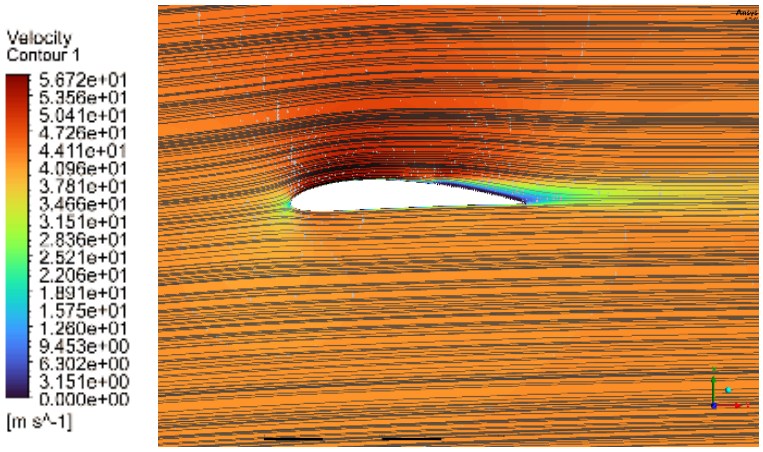


Figure 3.8 Velocity contour of 40% roughness airfoil at angle of attack 5

The 40% chord roughness introduces a localized disturbance as the boundary layer passes over the roughness element. Although this disturbance slightly modifies the velocity field in the vicinity of the roughness, the flow rapidly re-establishes itself downstream, resulting in a velocity distribution that remains comparable to that of the baseline configuration. The wake behind the roughness configuration appears only marginally more pronounced, suggesting that the disturbance is confined primarily to the local region rather than affecting the overall flow development.

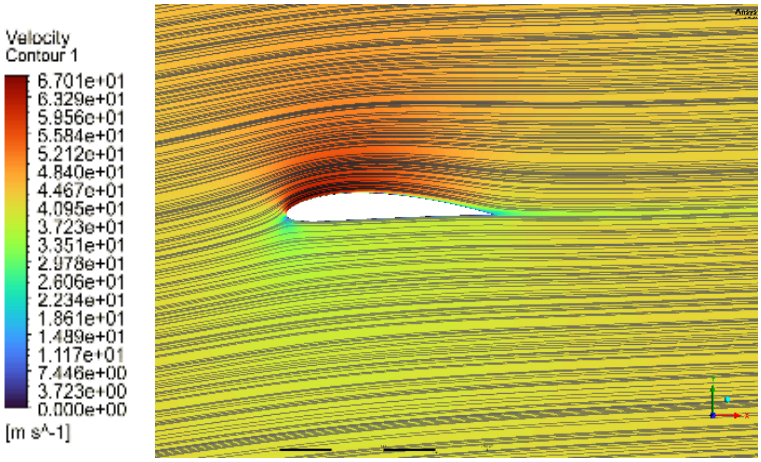


Figure 3.9 Velocity contour of baseline airfoil at angle of attack 10

At an angle of attack of 10°, the acceleration of the airflow over the upper surface increases further, producing a larger high-velocity region than that observed at lower angles. The baseline airfoil maintains a smooth velocity distribution along the suction surface, indicating that the boundary layer remains attached while generating the higher lift coefficient recorded in Section 3.1. The wake downstream of the trailing edge remains relatively narrow, suggesting limited aerodynamic losses under these operating conditions.

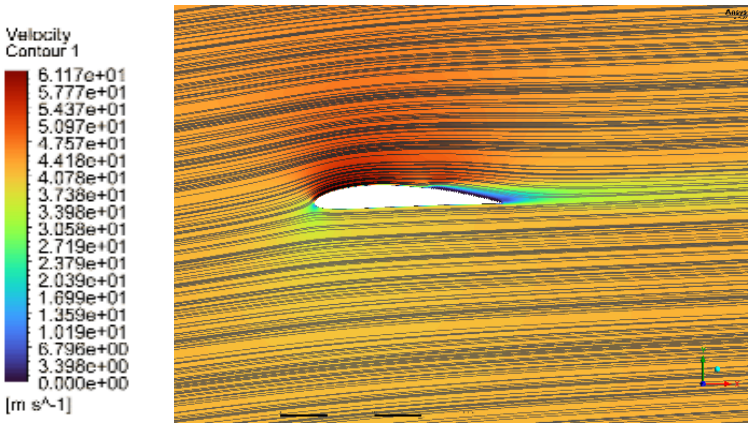


Figure 3.10 Velocity contour of 40% roughness airfoil at angle of attack 10

The velocity field changes slightly when the zig-zag roughness is introduced at 40% of the chord. As the boundary layer encounters the roughness element, a localized disturbance develops, producing a small reduction in the uniformity of the flow immediately downstream of the roughness. Although the boundary layer is affected locally, the flow remains attached over most of the airfoil surface, and no large-scale separation is evident. Compared with the baseline configuration, the wake behind the

roughness case appears marginally wider, indicating a modest increase in energy loss associated with the interaction between the flow and the surface irregularity.

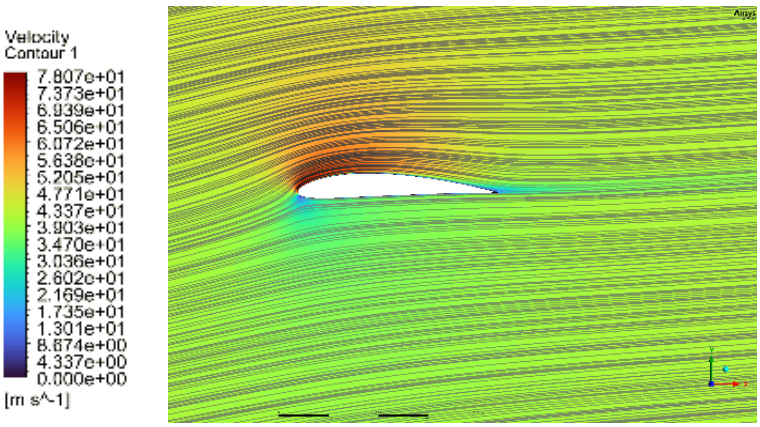


Figure 3.11 Velocity contour of baseline airfoil at angle of attack 15

At an angle of attack of 15°, the airflow reaches its highest level of acceleration over the upper surface before the onset of significant flow separation. The baseline airfoil exhibits an extensive high-velocity region along the suction surface, indicating a strong pressure difference across the airfoil and supporting the high lift coefficient obtained at this operating condition. Although the adverse pressure gradient becomes more pronounced near the trailing edge, the flow remains attached over most of the chord, resulting in only a limited wake downstream of the airfoil.

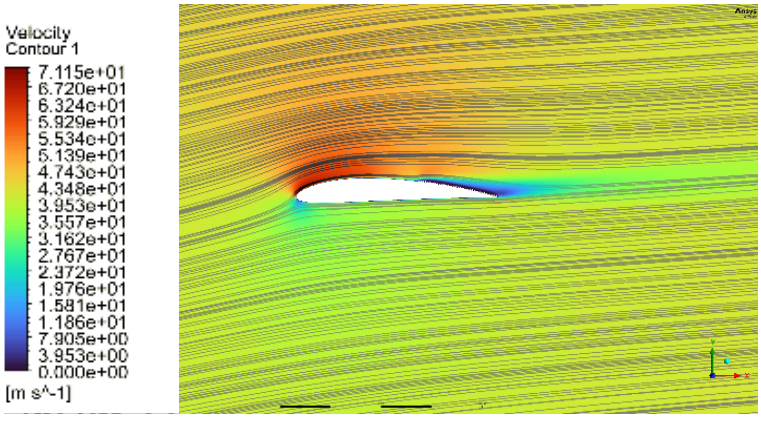


Figure 3.12 Velocity contour of 40% roughness airfoil at angle of attack 15

For the 40% chord roughness configuration, the interaction between the boundary layer and the roughness element becomes more noticeable than at the lower angles of attack. A localized disturbance develops immediately downstream of the roughness location, producing a slight redistribution of the velocity field along the upper surface. Even with this disturbance, the flow remains attached over the majority of the airfoil, suggesting that the roughness does not induce premature separation under

these conditions. The wake behind the roughness configuration appears marginally thicker than that of the baseline airfoil, indicating a small increase in aerodynamic losses associated with the modified boundary-layer development.

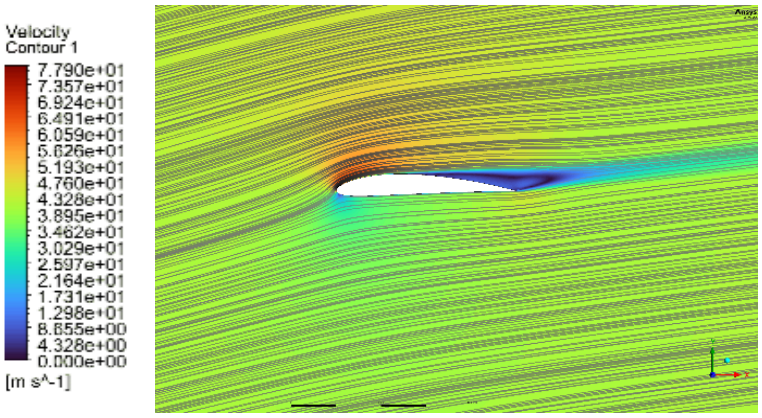


Figure 3.13 Velocity contour of baseline airfoil at angle of attack 20

At an angle of attack of 20°, the flow approaches the stall region, leading to a noticeable change in the velocity distribution around both airfoils. Compared with the 15° case, the high-velocity region over the upper surface becomes less extensive, indicating a reduction in the suction effect as the adverse pressure gradient increases. The downstream wake also becomes wider, reflecting greater momentum loss and the development of separated flow near the trailing portion of the airfoil.

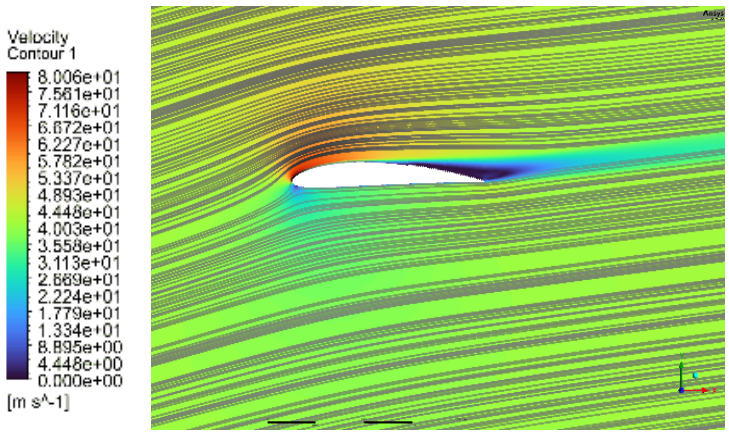


Figure 3.14 Velocity contour of 40% roughness airfoil at angle of attack 20

The influence of the zig-zag roughness becomes more evident under these conditions. As the boundary layer passes over the roughness element at 40% of the chord, the local velocity field is disturbed before continuing downstream. Despite this disturbance, the contour indicates that the flow remains relatively well organised over a significant portion of the upper surface, with no abrupt breakdown immediately after

the roughness location. The wake behind the roughness configuration differs slightly from that of the baseline, suggesting that the roughness modifies the boundary-layer development as the airfoil approaches stall.

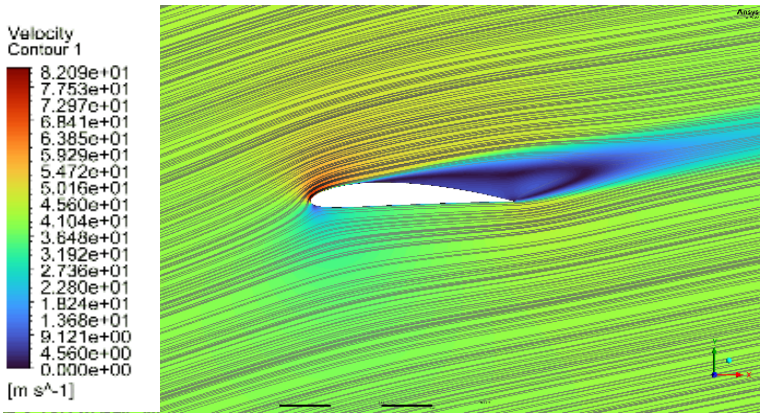


Figure 3.15 Velocity contour of baseline airfoil at angle of attack 25

At an angle of attack of 25°, both airfoils operate beyond the normal lifting range, and the velocity contours reflect the development of extensive flow separation. The high-velocity region over the suction surface is noticeably smaller than at 20°, indicating a substantial reduction in flow acceleration as the boundary layer loses its ability to remain fully attached. This change is accompanied by a significantly larger wake downstream of the airfoil, demonstrating increased momentum loss and reduced aerodynamic efficiency.

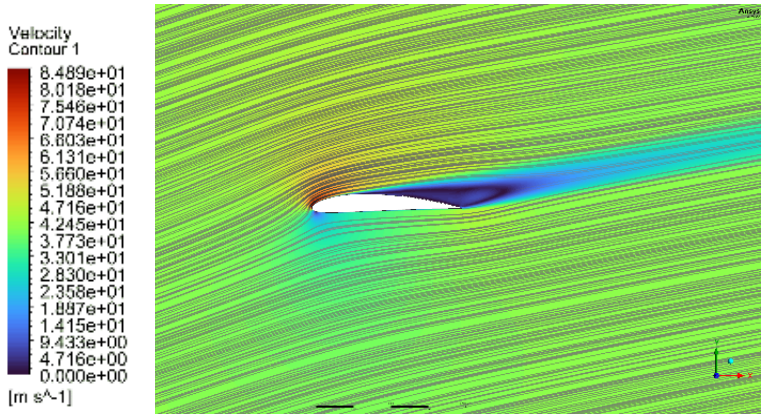


Figure 3.16 Velocity contour of 40% roughness airfoil at angle of attack 25

The 40% chord roughness configuration exhibits a similar overall flow pattern but with a modified velocity distribution in the vicinity of the roughness element. The disturbance introduced by the zig-zag roughness influences the development of the separated boundary layer, producing a downstream flow field that differs from the smooth baseline configuration. Although separation is present in both cases, the

roughness configuration maintains a more organised velocity distribution over part of the upper surface before the wake fully develops, suggesting that the roughness alters the progression of separation rather than eliminating it.

3.3 Analysis of Pressure Distribution

The pressure distribution around an airfoil directly influences the aerodynamic forces generated during flight by establishing the pressure difference between the upper and lower surfaces. Changes in the pressure field reflect the response of the boundary layer to increasing angle of attack and provide a clear indication of how surface roughness affects lift generation and the onset of flow separation. In this study, the pressure contours of the baseline NACA 4412 airfoil were compared with those of the 40% chord zig-zag roughness configuration under identical flow conditions. The following discussion examines the pressure distribution at each angle of attack to identify the influence of the roughness element on the aerodynamic characteristics of the airfoil.

The pressure contour analysis complements the velocity field by illustrating the pressure distribution responsible for lift generation. Relating the pressure field to the aerodynamic coefficients provides a better understanding of how roughness location influences the overall performance of the airfoil.

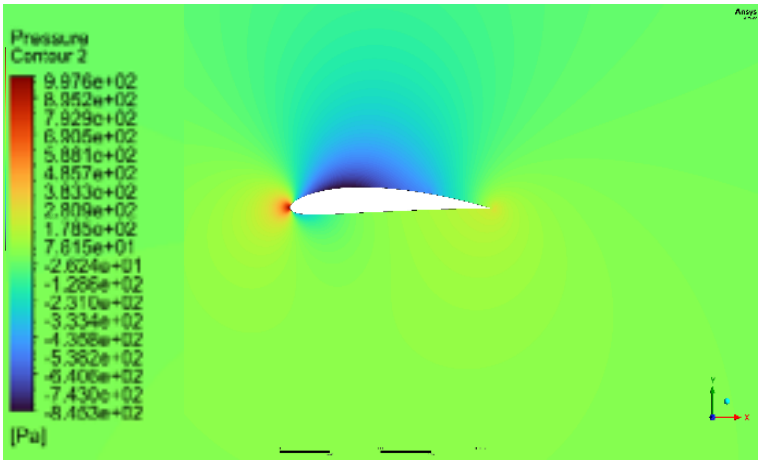


Figure 3.17 Pressure contour of baseline airfoil at angle of attack 0

At an angle of attack of 0°, the pressure distribution remains nearly symmetrical around both airfoils, with only a modest pressure difference established between the upper and lower surfaces. A region of higher pressure is formed near the leading edge where the incoming flow decelerates, while the upper surface experiences a slight reduction in pressure as the airflow accelerates over the cambered profile. Since the flow remains fully attached, the pressure recovery towards the trailing edge occurs gradually without any abrupt changes.

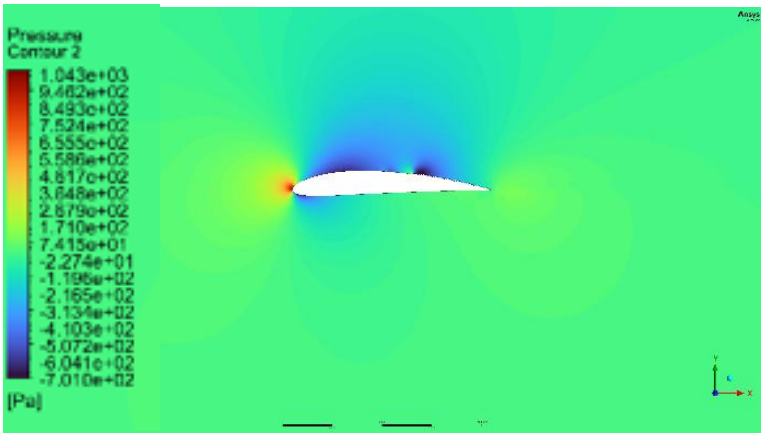


Figure 3.18 Pressure contour of 40% roughness airfoil at angle of attack 0

The introduction of the zig-zag roughness at 40% of the chord produces only a localized alteration in the pressure field. A small disturbance develops around the roughness element; however, the overall pressure distribution remains comparable to that of the baseline configuration. The suction region over the upper surface is preserved, indicating that the roughness does not significantly influence pressure development while the boundary layer remains stable.

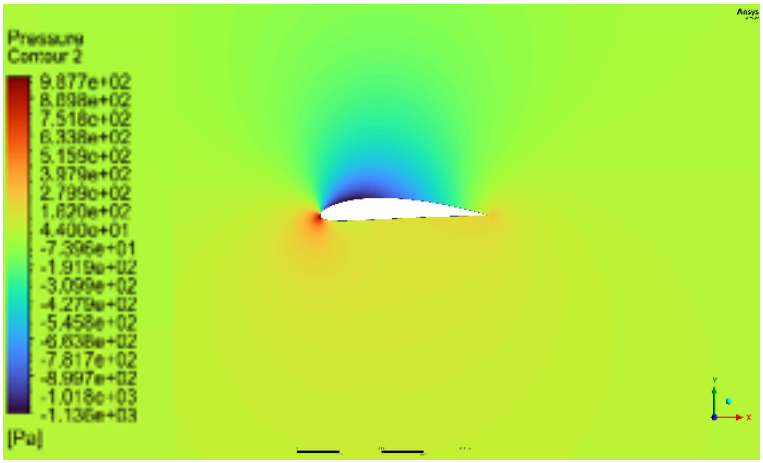


Figure 3.19 Pressure contour of baseline airfoil at angle of attack 5

At an angle of attack of 5°, the pressure distribution becomes more asymmetric than at 0°, reflecting the increase in lift generated by the airfoil. The pressure on the upper surface decreases further as the airflow accelerates over the suction side, while the lower surface maintains a comparatively higher pressure. This increase in pressure difference enhances the lifting capability of the baseline airfoil without producing any indication of significant pressure loss or premature flow separation.

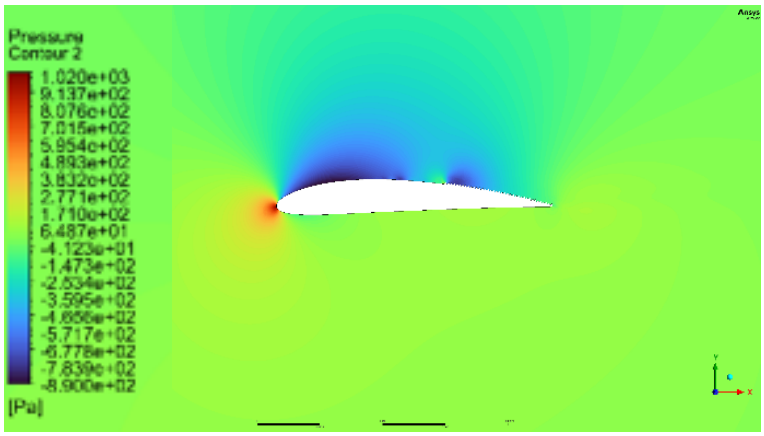


Figure 3.20 Pressure contour of 40% roughness airfoil at angle of attack 5

The pressure field around the 40% chord roughness configuration follows a similar overall pattern. A localized variation in pressure is observed in the vicinity of the roughness element as the boundary layer interacts with the surface disturbance. However, the low-pressure region over the upper surface remains largely continuous, and the pressure recovery towards the trailing edge occurs smoothly. These characteristics indicate that the roughness does not significantly disturb the overall pressure distribution under moderate operating conditions.

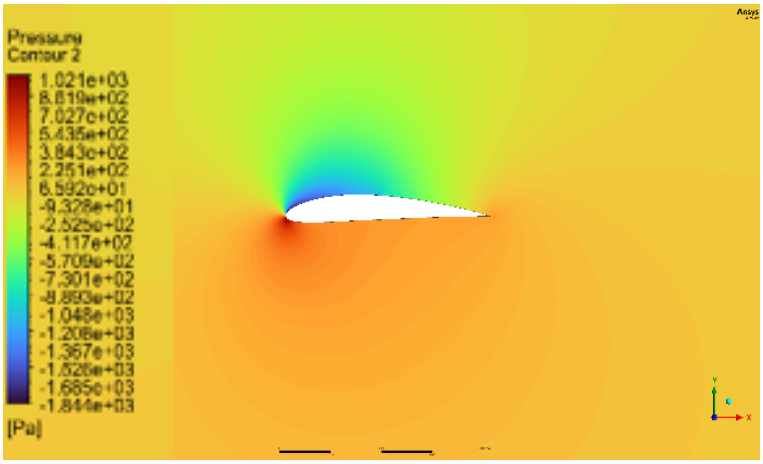


Figure 3.21 Pressure contour of baseline airfoil at angle of attack 10

At an angle of attack of 10°, the pressure difference across the airfoil increases further as the low-pressure region over the suction surface expands. The baseline airfoil exhibits a well-defined suction region extending over a large portion of the upper surface, while the pressure on the lower surface remains comparatively higher. This pressure distribution promotes increased lift generation and reflects the favourable aerodynamic performance observed at this operating condition.

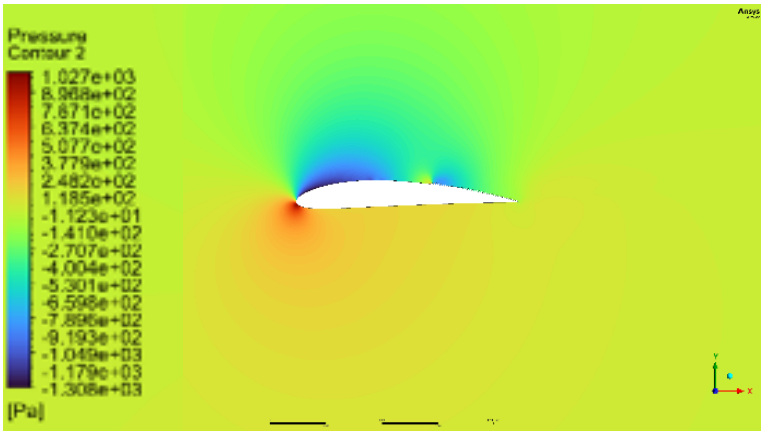


Figure 3.22 Pressure contour of 40% roughness airfoil at angle of attack 10

For the 40% chord roughness configuration, the pressure contours display a similar overall distribution, although a localized variation is evident near the roughness element where the boundary layer interacts with the surface feature. Despite this disturbance, the suction region remains continuous along most of the upper surface, and the pressure recovery towards the trailing edge occurs without any abrupt changes. This indicates that the roughness modifies the local pressure field without significantly affecting the overall pressure distribution around the airfoil.

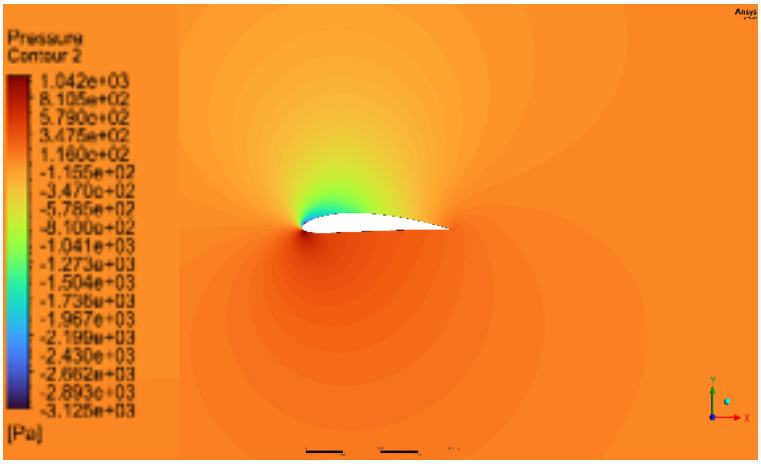


Figure 3.23 Pressure contour of baseline airfoil at angle of attack 15

At an angle of attack of 15°, the pressure contours exhibit the largest pressure difference between the upper and lower surfaces among the pre-stall operating conditions. The baseline airfoil develops an extensive low-pressure region over the suction surface while maintaining comparatively higher pressure along the lower surface. This pressure distribution promotes strong lift generation and corresponds to the high lift coefficient obtained for the baseline configuration in Section 3.1.

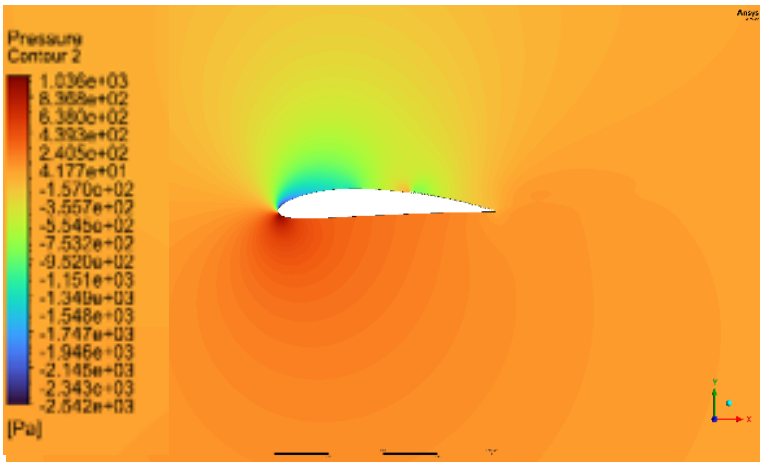


Figure 3.24 Pressure contour of 40% airfoil at angle of attack 15

The pressure distribution for the 40% chord roughness configuration follows a similar pattern, with the roughness introducing only a localized disturbance near its position on the upper surface. The suction region remains largely uninterrupted over most of the chord, indicating that the boundary layer continues to sustain the pressure difference required for effective lift generation. Pressure recovery towards the trailing edge occurs gradually in both configurations, suggesting that large-scale flow separation has not yet become dominant at this angle of attack.

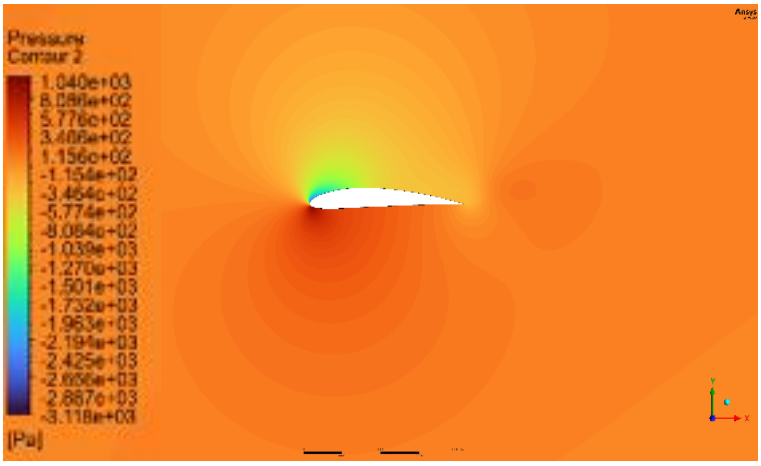


Figure 3.25 Pressure contour of baseline airfoil at angle of attack 20

At an angle of attack of 20°, the pressure distribution begins to reflect the onset of stall. Compared with the 15° condition, the low-pressure region over the upper surface becomes less extensive, indicating a reduction in the suction force responsible for lift generation. At the same time, pressure recovery towards the trailing edge becomes more gradual, suggesting that the adverse pressure gradient is becoming increasingly influential and the boundary layer is approaching separation.

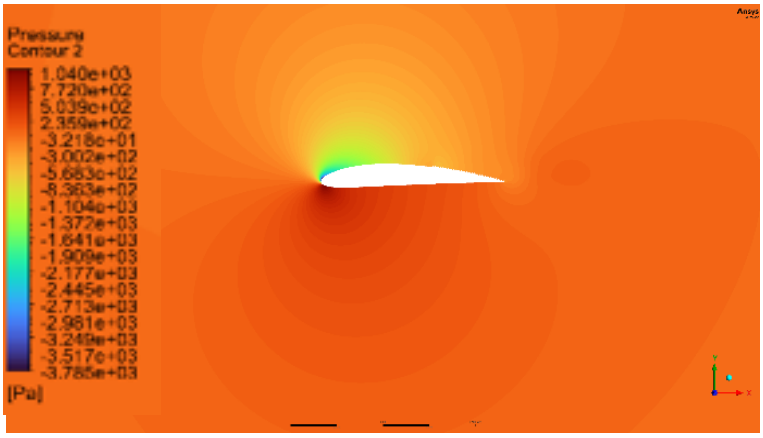


Figure 3.26 Pressure contour at 40% roughness at angle of attack 20

The pressure contours of the 40% chord roughness configuration exhibit a similar overall pattern, although a localized variation is visible around the roughness element where the flow interacts with the surface feature. Despite this local disturbance, the suction region remains reasonably well established over a significant portion of the upper surface, and the pressure distribution does not undergo any abrupt breakdown. Compared with the baseline airfoil, the downstream pressure field shows only minor differences, indicating that the roughness primarily modifies the local pressure characteristics rather than the overall loading on the airfoil.

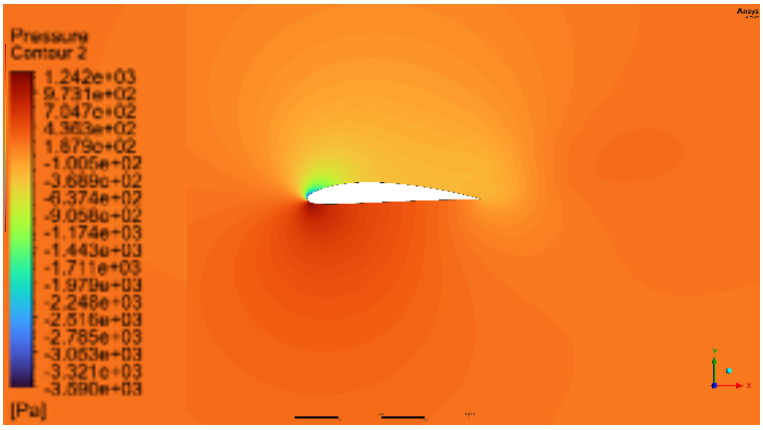


Figure 3.27 Pressure contour of baseline airfoil at angle of attack 25

At an angle of attack of 25°, the pressure distribution indicates that both airfoils have entered the post-stall region. The low-pressure region over the suction surface is considerably smaller than that observed at 20°, demonstrating a reduction in the pressure difference responsible for lift generation. As the separated flow region expands, the pressure recovery towards the trailing edge becomes less effective, resulting in a more uniform pressure distribution over the upper surface and a corresponding decline in aerodynamic performance.

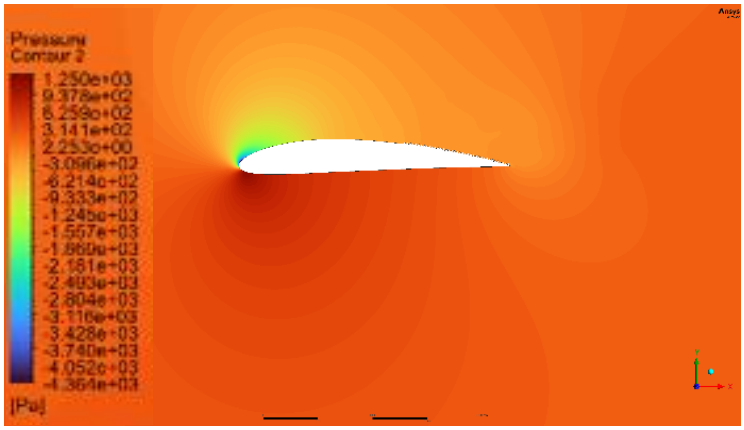


Figure 3.28 Pressure contour of 40% roughness at angle of attack 25

The pressure contours of the 40% chord roughness configuration display a similar overall trend, although the pressure field around the roughness element differs slightly from that of the baseline airfoil. A localized modification in the pressure distribution is evident near the roughness location, while the suction region remains identifiable over part of the upper surface before gradually weakening towards the trailing edge. Compared with the baseline configuration, the pressure recovery appears

to occur more progressively, suggesting that the roughness influences the development of the separated boundary layer during the post-stall condition.

3.4 Lift-to-Drag Ratio Analysis

The lift-to-drag ratio provides an overall measure of aerodynamic efficiency by relating the lift generated to the drag experienced by the airfoil. Figure 3.15 compares the variation of the lift-to-drag ratio for the baseline configuration and all zig-zag roughness locations over the investigated range of angles of attack.

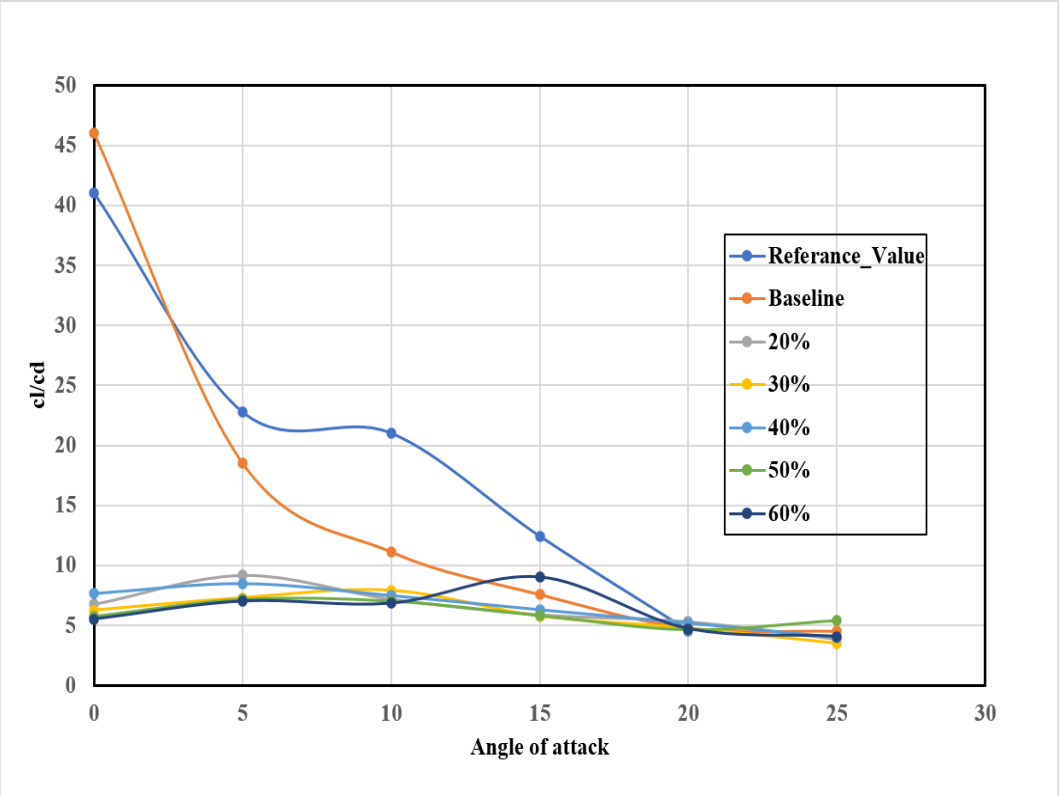


Figure 3.29 Variation of lift to drag ratio (cl/cd) with angle of attack for the baseline airfoil and all roughness configuration

All configurations exhibit a similar overall trend, with the lift-to-drag ratio reaching its highest values at lower angles of attack before decreasing progressively as the angle of attack increases. This behaviour reflects the gradual shift from an attached flow regime, where lift increases more rapidly than drag, to a condition in which the growth of drag becomes increasingly dominant. Beyond the moderate operating range, the adverse pressure gradient strengthens and flow separation develops over the upper surface, leading to a noticeable reduction in aerodynamic efficiency.

The influence of roughness location becomes more apparent as the angle of attack increases. At lower incidence, only minor differences are observed between the configurations, indicating that the boundary layer remains sufficiently stable to limit the effect of the surface disturbance. As the airfoil approaches the stall region, however, the aerodynamic efficiency of each configuration begins to diverge. This behaviour indicates that the location of the roughness element influences the manner in which the boundary layer develops under increasing adverse pressure gradients.

Among the roughness configurations investigated, the 40% chord location maintains a favourable balance between lift generation and drag over a wide range of operating conditions. The velocity and pressure contour analyses support this observation by showing that the roughness introduces only a localized disturbance while preserving an effective pressure distribution over much of the upper surface. Consequently, the airfoil retains useful lift without a proportional increase in drag, resulting in comparatively higher aerodynamic efficiency than several of the other roughness locations.

The reduction in the lift-to-drag ratio at 20° and 25° is consistent with the flow-field observations presented in the previous sections. At these higher angles of attack, the expansion of the separated flow region weakens the suction effect over the upper surface and increases aerodynamic losses, causing the ratio to decline for all configurations. Even under these conditions, the 40% chord roughness continues to demonstrate competitive aerodynamic performance, supporting its selection for the detailed contour comparison carried out in this study.

Overall, the lift-to-drag ratio analysis reinforces the conclusion that the chordwise location of the zig-zag roughness plays an important role in determining aerodynamic efficiency. Considering the aerodynamic coefficients together with the pressure and velocity contour analyses, the 40% chord roughness configuration provides the most balanced aerodynamic performance among the configurations investigated.

CHAPTER 4
CONCLUSION

The present work investigated the effect of zig-zag surface roughness on the aerodynamic performance of the NACA 4412 airfoil using Computational Fluid Dynamics (CFD). The primary objective was to examine whether the placement of surface roughness could influence boundary-layer behaviour, reduce the effects of flow separation at higher angles of attack, and improve the overall aerodynamic performance of the airfoil. To achieve this, a smooth baseline airfoil and five roughness configurations positioned at 20%, 30%, 40%, 50% and 60% of the chord length were analysed under identical flow conditions for angles of attack ranging from 0° to 25°.

The aerodynamic characteristics of each configuration were evaluated using the lift coefficient, drag coefficient, lift-to-drag ratio, and the corresponding velocity and pressure contours. Comparing these parameters made it possible to relate the changes in aerodynamic performance to the flow behaviour around the airfoil. At lower angles of attack, the flow remained attached over all configurations, resulting in only small differences in lift and drag. As the angle of attack increased, the influence of roughness location became more evident because of changes in boundary-layer development, pressure distribution and the gradual onset of flow separation.

Among the roughness locations investigated, the configuration with the zig-zag roughness placed at 40% of the chord consistently produced the most balanced aerodynamic performance. The velocity contours showed that the disturbance created by the roughness remained localised without causing premature flow breakdown, while the pressure contours indicated that the pressure difference between the upper and lower surfaces was largely maintained even at higher angles of attack. These flow characteristics contributed to favourable lift generation without a corresponding increase in drag, leading to improved aerodynamic efficiency compared with the other roughness locations.

As the airfoil entered the stall region, all configurations experienced a reduction in aerodynamic efficiency because of the expansion of the separated flow region and the accompanying increase in drag. Even under these conditions, the 40% roughness configuration retained comparatively better aerodynamic characteristics than the

remaining roughness positions. This indicates that placing the zig-zag roughness near the mid-chord region modifies the boundary-layer development in a manner that helps reduce the adverse effects of flow separation while preserving useful aerodynamic loading.

From the conditions investigated in this study, the 40% chord roughness configuration is identified as the most effective arrangement for the NACA 4412 airfoil. The results demonstrate that the aerodynamic performance of an airfoil is influenced not only by the presence of surface roughness but also by its location along the chord. The findings of this work provide useful guidance for the placement of artificial roughness elements in aerodynamic applications where maintaining lift and delaying the effects of flow separation are important considerations.

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